



200 Logic Building Questions

6 Phases · Carefully curated from scratch · Master these before you tackle DSA

P1

PHASE 1 — CONDITIONAL THINKING

Goal: Understand how to make decisions using conditions

Topics: Relational operators, logical operators, nested if, compound conditions

30 Qs

● Level 1 — Basic Conditions (Q 1–10)

- 1 Take a number and print whether it is Positive, Negative, or Zero.
- 2 Check if a given number is Even or Odd.
- 3 Check if a number is exactly divisible by 5.
- 4 Check if a number is divisible by both 3 and 5 simultaneously.
- 5 Given a year, determine whether it is a Leap Year or not.
- 6 Take two numbers and print the Larger one.
- 7 Take three numbers and print the Largest among them.
- 8 Take a character and check if it is a Vowel or a Consonant.
- 9 Take a character and classify it as Uppercase letter, Lowercase letter, Digit, or Special character.
- 10 Check if a person is eligible to Vote given their age (18+ = eligible).

● Level 2 — Intermediate Conditions (Q 11–20)

- 11 Take three sides as input and check whether they can form a Valid Triangle.
- 12 If the triangle is valid, determine its Type: Equilateral, Isosceles, or Scalene.
- 13 Take marks from 0–100 and print the corresponding Grade (A / B / C / D / F).
- 14 Take a Day number (1–7) and print the corresponding Day name (Monday–Sunday).
- 15 Take a Month number (1–12) and print the number of Days in that month (ignore leap years).
- 16 Check if a given integer is Single-digit, Double-digit, or Multi-digit.
- 17 Check whether a given number lies within the range [100, 999].
- 18 Take coordinates (x, y) and determine which Quadrant the point lies in, or if it's on an axis.
- 19 Check whether a given positive integer is a Perfect Square without using the sqrt function.
- 20 Given three numbers, print the Median value (neither the maximum nor the minimum).

● Level 3 — Advanced Logic (Q 21–30)

- 21 Given electricity units consumed, calculate the Bill amount using predefined tariff slabs with if-else.
- 22 Check basic Password Validity: length must be at least 8 characters and contain at least one digit.



- 23 Take a day and month and check if it forms a Valid Calendar Date (ignore leap years).
- 24 Given a year, print the corresponding Century (e.g., 2001–2100 → '21st Century').
- 25 Take three integers and check if they can form a Pythagorean Triplet ($a^2+b^2=c^2$).
- 26 Take three numbers and check if they are in Arithmetic Progression (constant common difference).
- 27 Take three numbers and check if they are in Geometric Progression (constant common ratio).
- 28 Given time in hours and minutes, compute the smaller Angle between the clock's hour and minute hands.
- 29 Take two dates (day and month) and determine which Date comes First in the calendar year.
- 30 Check Tax Eligibility: a person is taxable if age > 18 and annual income exceeds 5 lakhs.

P2

PHASE 2 — LOOPS + NUMBER THINKING

Goal: Master loops, iteration, and dry-run thinking

Topics: for, while, nested loops, break/continue, mathematical series

45 Qs

● Level 1 — Basic Loops (Q 31–40)

- 31 Print all integers from 1 to n using a loop.
- 32 Print integers in reverse from n down to 1.
- 33 Print all Even numbers from 1 up to n.
- 34 Print all Odd numbers from 1 up to n.
- 35 Print the Multiplication Table of a given number n ($n \times 1$ through $n \times 10$).
- 36 Compute and print the Sum of the first n natural numbers.
- 37 Compute and print the Sum of all Even numbers from 1 up to n.
- 38 Compute and print the Sum of all Odd numbers from 1 up to n.
- 39 Calculate and print the Factorial of a given number n.
- 40 Compute and print the Product of all Digits of a given number.

● Level 2 — Number Logic (Q 41–55)

- 41 Count and print the total number of Digits in a given integer.
- 42 Reverse a given integer and print the result (e.g., 1234 → 4321).
- 43 Check whether a given number is a Palindrome (reads the same forwards and backwards).
- 44 Calculate and print the Sum of all Digits of a given number.
- 45 Check whether a number is an Armstrong Number (sum of digits each raised to the power of digit count equals the number).
- 46 Check whether a number is a Perfect Number (sum of its proper divisors equals itself).
- 47 Check if a given number is Prime or Not.
- 48 Print all Prime Numbers from 1 to n.



- 49 Print the Fibonacci Series up to n terms.
- 50 Calculate and print the Sum of the first n Fibonacci terms.
- 51 Print the Squares of all numbers from 1 to n.
- 52 Print the Cubes of all numbers from 1 to n.
- 53 Find and print the HCF (GCD) of two numbers using a loop.
- 54 Find and print the LCM of two numbers using a loop.
- 55 Print all Factors of a given number, then compute their sum.

● Level 3 — Strong Number Logic (Q 56–65)

- 56 Check whether a number is a Strong Number: the sum of the factorials of its digits equals the number itself.
- 57 Print the first n terms of an Arithmetic Progression given the first term 'a' and common difference 'd'.
- 58 Print the first n terms of a Geometric Progression given the first term 'a' and common ratio 'r'.
- 59 Print all numbers from 1 to 100 whose Sum of Digits is Even.
- 60 Count how many numbers between 1 and 500 are Divisible by 7 but not by 5.
- 61 Print all Palindrome Numbers between 1 and 500.
- 62 Print all numbers from 1 to 100 whose digit sum is a Multiple of 3.
- 63 Find and print the Smallest Digit and Largest Digit present in a given number.
- 64 Print all numbers from 1 to n whose Binary Representation contains an Even count of 1s.
- 65 Print the Factorial of every number from 1 to n (one per line).

● Level 4 — Mixed Loop Logic (Q 66–70)

- 66 For a given number, separately compute the Sum of its Odd Digits and the Sum of its Even Digits.
- 67 Take 5 numbers as input; use the continue statement to Skip any zero entries and print the sum of all non-zero numbers.
- 68 Print all Pairs from a given array whose Sum equals a target value k.
- 69 Print all Subarrays of a given array along with each subarray's sum.
- 70 Print a 10×10 Multiplication Grid with proper formatting.

● Level 5 — Advanced Dry Run (Q 71–75)

- 71 Count how many times a particular number appears Consecutively in an array (e.g., [1,1,2,1] → max 2 for 1).
- 72 Print Pascal's Triangle up to n rows using nested loops.
- 73 Given a number, count how many numbers from 1 to n have a Digit Sum divisible by that number.



74 Print all numbers from 1 to n in a format where each row i prints i copies of i (triangular pattern).

75 Simulate printing numbers in a Zigzag pattern across n rows.

P3

PHASE 3 — PATTERNS

Goal: Develop nested loop thinking through visual pattern printing

Topics: Nested loops, star patterns, number patterns, alphabet patterns

35 Qs

Level 1 — Star Basics (Q 76–85)

- 76 Print a single star (*).
- 77 Print four stars (****) in a single row.
- 78 Print n stars in a single row where n is given by the user.
- 79 Print a Solid Square of stars of size $n \times n$.
- 80 Print a Right-Angled Triangle of stars with increasing stars per row (1, 2, 3 ... n).
- 81 Print a Right-Aligned Triangle where each row is right-justified.
- 82 Print rows of stars only for Even row numbers up to n.
- 83 Print rows of stars only for Odd row numbers up to n.
- 84 Print a Pyramid pattern of stars centered on each row.
- 85 Print a Hollow Pyramid — only the boundary stars, inside is empty.

Level 2 — Advanced Stars (Q 86–95)

- 86 Print a Reverse Triangle (decreasing stars from n to 1).
- 87 Print a Diamond pattern (upper pyramid + lower reverse pyramid).
- 88 Print a Full Diamond that includes both upper and lower halves with equal width.
- 89 Print an Hourglass pattern (reverse pyramid on top, pyramid on bottom).
- 90 Print a Hollow Square — only border stars, interior is blank.
- 91 Print an X Pattern using stars across two diagonals.
- 92 Print a Butterfly Pattern (two mirrored triangles per row).
- 93 Print a Cross Pattern (horizontal and vertical center lines filled with stars).
- 94 Print a Sandglass Pattern (hourglass touching at a single point in the middle).
- 95 Print a Staircase Pattern (one extra star each new step, like ascending stairs).

Level 3 — Number Patterns (Q 96–105)

- 96 Print a triangle where row i contains the number i repeated i times.
- 97 Print a triangle where each row r shows 1, 2, 3, ... r.
- 98 Print a triangle where each row shows 1, 1 2, 1 2 3 ... (Floyd's Triangle style).
- 99 Print Floyd's Triangle: continuous numbers filled row by row.



- 100 Print a triangle where each row repeats its row number (e.g., row 3 → 3 3 3).
- 101 Print a number triangle where digits continue sequentially across all rows.
- 102 Print a Binary Triangle alternating 0 and 1 in each row.
- 103 Print a Palindrome Number Pyramid (e.g., row 3 → 1 2 3 2 1).
- 104 Print a Centered Number Pyramid with numbers increasing to the peak then decreasing.
- 105 Print a Zigzag Numbers pattern alternating between ascending and descending rows.

● Level 4 — Alphabet Patterns (Q 106–110)

- 106 Print a right-angled triangle of alphabets: row 1 → A, row 2 → A B, row 3 → A B C, etc.
- 107 Print an Alphabet Pyramid centered, with letters increasing to the middle column.
- 108 Print a pattern where each row r repeats the letter corresponding to r (row 1 → A, row 2 → B B, etc.).
- 109 Print all 26 letters A–Z each on its own line.
- 110 Print the first n letters of the alphabet, each on its own line.

P4

PHASE 4 — RECURSION

Goal: Develop logical decomposition and base-condition thinking

Topics: Recursive definition, base cases, call stack tracing

30 Qs

● Level 1 — Basic Recursion (Q 111–120)

- 111 Print all numbers from 1 to n using Recursion (no loops allowed).
- 112 Print all numbers from n down to 1 using Recursion.
- 113 Print only Even numbers from 1 to n using Recursion.
- 114 Print only Odd numbers from 1 to n using Recursion.
- 115 Compute the Sum of the first n natural numbers using Recursion.
- 116 Compute the Factorial of n using Recursion.
- 117 Compute x raised to the power n (x^n) using Recursion.
- 118 Find the nth Fibonacci number using Recursion.
- 119 Print the entire Fibonacci Series up to n terms using Recursion.
- 120 Find the Sum of Digits of a number using Recursion.

● Level 2 — Number Recursive Logic (Q 121–130)

- 121 Count the number of Digits in a number using Recursion.
- 122 Reverse a given number using Recursion and print the result.
- 123 Check if a number is a Palindrome using Recursion.
- 124 Find the Product of all Digits of a number using Recursion.
- 125 Find the GCD (HCF) of two numbers using Euclid's Algorithm recursively.



- 126 Convert a Decimal number to its Binary representation using Recursion.
- 127 Print each Digit of a number as a Word recursively (e.g., 123 → 'one two three').
- 128 Calculate the Sum of the first n Even numbers using Recursion.
- 129 Calculate the Sum of the first n Odd numbers using Recursion.
- 130 Compute nCr (combination) using Pascal's Relation recursively.

● Level 3 — String Recursion (Q 131–140)

- 131 Reverse a String using Recursion without using any built-in reverse function.
- 132 Check whether a String is a Palindrome using Recursion.
- 133 Count the number of Vowels in a string using Recursion.
- 134 Remove all Spaces from a string using Recursion.
- 135 Replace every occurrence of character 'a' with 'x' in a string using Recursion.
- 136 Remove all occurrences of a given character from a string using Recursion.
- 137 Print each Character of a string one by one using Recursion (no loops).
- 138 Print a string in Reverse Order using Recursion (no loops, no built-ins).
- 139 Convert all characters of a string to Uppercase using Recursion.
- 140 Count Vowels and Consonants separately in a string using Recursion.

P5

PHASE 5 — ARRAYS

Goal: Build the ability to handle a collection of data logically
Topics: Traversal, frequency, simple manipulation, aggregations

40 Qs

● Level 1 — Array Fundamentals (Q 141–150)

- 141 Take n integers as input into an array and print them.
- 142 Find and print the Sum of all elements in an array.
- 143 Calculate and print the Average of all array elements.
- 144 Find and print the Maximum element in an array.
- 145 Find and print the Minimum element in an array.
- 146 Count how many elements in an array are Positive, Negative, or Zero.
- 147 Count how many elements in an array are Even and how many are Odd.
- 148 Find and print the Index of the Maximum element.
- 149 Find and print the Index of the Minimum element.
- 150 Print only those elements that are greater than a given threshold value k.

● Level 2 — Searching & Counting (Q 151–160)

- 151 Check if a given element x Exists in the array.



152 Count how many times a given element appears (its Frequency).

153 Find the Index of the First Occurrence of a given number.

154 Find the Index of the Last Occurrence of a given number.

155 Check if All Elements in the array are Unique (no duplicates).

156 Find the Sum of Even Elements only.

157 Find the Sum of Odd Elements only.

158 Count how many elements in the array are Prime Numbers.

159 Count elements Divisible by both 3 and 5.

160 Count how many elements are Perfect Squares.

● Level 3 — Transformation & Manipulation (Q 161–170)

161 Create a new array containing the Squares of all original elements.

162 Create a new array containing only the Even Elements from the original.

163 Replace every Negative Number in the array with 0.

164 Replace all Even numbers with 1 and all Odd numbers with 0 in-place.

165 Swap the First and Last elements of the array.

166 Reverse an array In-Place without using any built-in reverse function.

167 Left Rotate the array by one position (first element goes to the end).

168 Right Rotate the array by one position (last element goes to the front).

169 Swap Alternate Elements (1st↔2nd, 3rd↔4th, etc.) in the array.

170 Copy one array into another array Manually element by element.

● Level 4 — Comparative & Strong Logic (Q 171–180)

171 Compare two arrays — check if they are Equal (same elements in same order).

172 Compare two arrays — check if they contain the Same Elements ignoring order.

173 Merge two arrays into a third array.

174 Find the Common Elements between two arrays.

175 Find elements that appear in one array but Not in the other.

176 Build a Frequency Array to count how many times each number (0–9) appears.

177 Print all elements that appear More Than Once (duplicates).

178 Check if the array is Sorted in Ascending or Descending order.

179 Find the Second Largest and Second Smallest elements.

180 Count the number of Pairs whose sum equals a given value k.



P6

PHASE 6 — STRINGS

Goal: Strengthen logical thinking with string and character manipulation

Topics: String traversal, character logic, word-level operations, palindromes

20 Qs

● Level 1 — Basic String Handling (Q 181–190)

- 181 Take a string as input and print its Length.
- 182 Print the First and Last characters of a string.
- 183 Convert and print the entire string in Uppercase.
- 184 Convert and print the entire string in Lowercase.
- 185 Count and print the number of Characters in the string excluding spaces.
- 186 Count and print how many Words are in a given sentence.
- 187 Take two strings and print their Concatenation.
- 188 Compare two strings Lexicographically and state which comes first.
- 189 Print the ASCII value of every Character in a string.
- 190 Check whether the given string is Empty or not.

● Level 2 — Important String Logic (Q 191–200)

- 191 Count the number of Vowels and Consonants in a string.
- 192 Count Digits, Letters, and Special Characters separately in a string.
- 193 Find the Frequency of each character in a string without using a map/dictionary.
- 194 Reverse a string without using any built-in Reverse function.
- 195 Reverse each Individual Word in a sentence while keeping word order intact.
- 196 Reverse the order of Words in a sentence (first word becomes last, etc.).
- 197 Check whether a string is a Palindrome.
- 198 Remove all Vowels from a string and print the result.
- 199 Remove Duplicate characters from a string, keeping only the first occurrence of each.
- 200 Remove Extra Spaces from a sentence and normalize it to single spaces between words.

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